



Quantigration RMA Data Analysis Report

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RMA Report

Write a report to respond to the manager's requests. In the report, you should complete the following actions:

- Summarize the data you've been working with.
- Identify key information that will help the company streamline operations.

Your report should explain your findings in a way nontechnical stakeholders can understand and use.

Use the steps below to capture the required data and produce the analysis report.

1. Begin by writing SQL commands to **capture** specific **usable data** for your analysis. You already preloaded the data you need into Codio.
2. Specifically, the product manager wants you to complete the following analysis:

A. Analyze the number of returns by state and describe findings to include in your report.

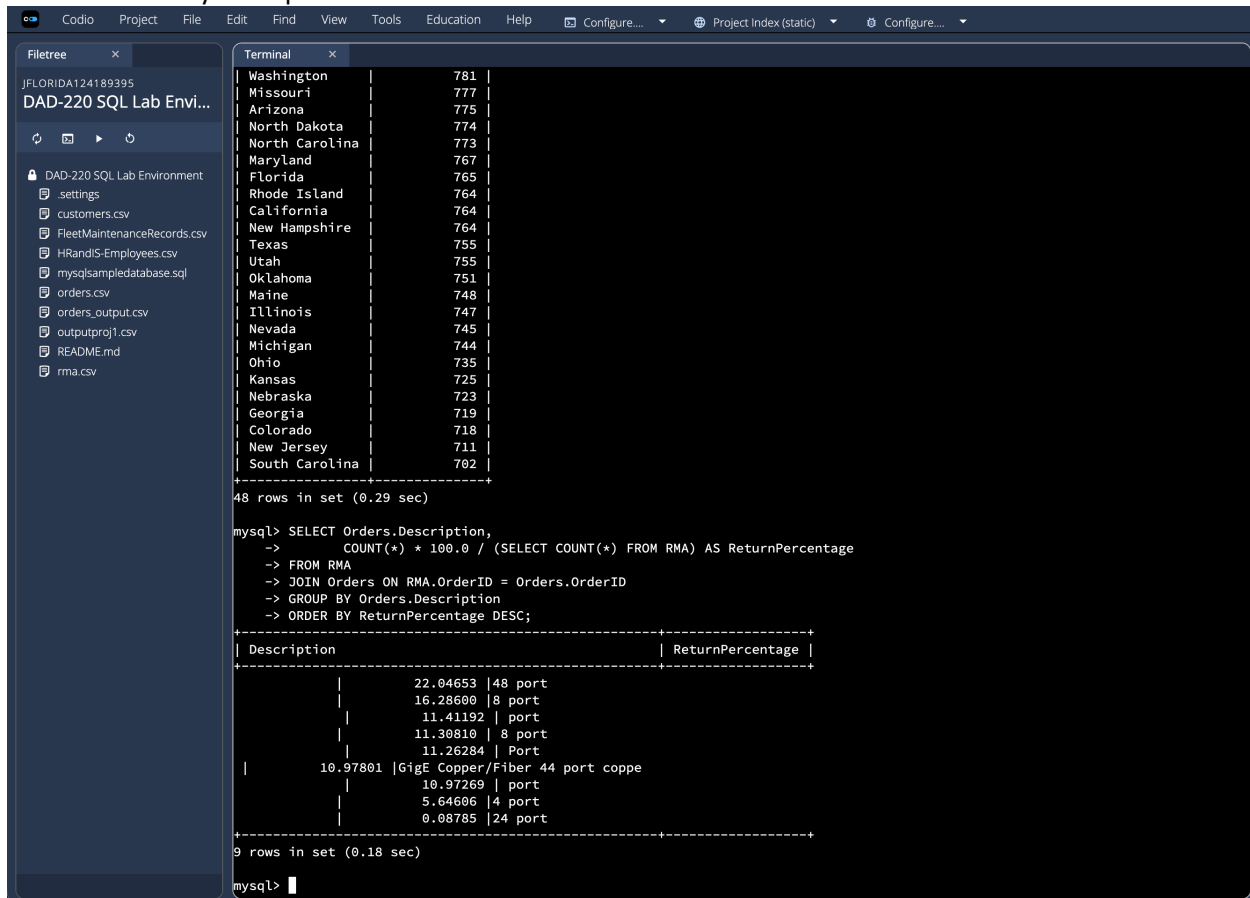
```
mysql> SELECT Customers.State, COUNT(*) AS TotalReturns
-> FROM RMA
-> JOIN Orders ON RMA.OrderID = Orders.OrderID
-> JOIN Customers ON Orders.CustomerID = Customers.CustomerID
-> GROUP BY Customers.State
-> ORDER BY TotalReturns DESC;
```

State	TotalReturns
Massachusetts	972
Arkansas	844
Oregon	840
West Virginia	837
Alabama	836
Connecticut	822
Idaho	822
Mississippi	821
Tennessee	819
Delaware	811
Kentucky	809
Montana	808
Wisconsin	807
New Mexico	807
Iowa	804
Indiana	802
Pennsylvania	802
South Dakota	797
Louisiana	794
Minnesota	794
Wyoming	786
Vermont	785
Hawaii	783
New York	782
Washington	781
Missouri	777
Arizona	775
North Dakota	774
North Carolina	773
Maryland	767
Florida	765
Rhode Island	764
California	764
New Hampshire	764
Texas	755
Utah	755
Oklahoma	751
Maine	748
Illinois	747
Nevada	745
Michigan	744
Ohio	735
Kansas	725
Nebraska	723
Georgia	719
Colorado	718
New Jersey	711
South Carolina	702

48 rows in set (0.29 sec)

This query analyzes the number of returns by state by joining the RMA, Orders, and Customers tables. It groups the data by state and counts the total number of returns for each state. The results show which states have the highest return activity, with Massachusetts having the highest number of returns, followed by Arkansas and Oregon.

B. Analyze the percentage of returns by product type and describe findings to include in your report.



```

Washington | 781 |
Missouri | 777 |
Arizona | 775 |
North Dakota | 774 |
North Carolina | 773 |
Maryland | 767 |
Florida | 765 |
Rhode Island | 764 |
California | 764 |
New Hampshire | 764 |
Texas | 755 |
Utah | 755 |
Oklahoma | 751 |
Maine | 748 |
Illinois | 747 |
Nevada | 745 |
Michigan | 744 |
Ohio | 735 |
Kansas | 725 |
Nebraska | 723 |
Georgia | 719 |
Colorado | 718 |
New Jersey | 711 |
South Carolina | 702 |
+-----+
48 rows in set (0.29 sec)

mysql> SELECT Orders.Description,
-> COUNT(*) * 100.0 / (SELECT COUNT(*) FROM RMA) AS ReturnPercentage
-> FROM RMA
-> JOIN Orders ON RMA.OrderID = Orders.OrderID
-> GROUP BY Orders.Description
-> ORDER BY ReturnPercentage DESC;
+-----+
| Description | ReturnPercentage |
+-----+
| 22.04653 | 48 port |
| 16.28600 | 8 port |
| 11.41192 | port |
| 11.30810 | 8 port |
| 11.26284 | Port |
| 10.97801 | GigE Copper/Fiber 44 port coppe |
| 10.97269 | port |
| 5.64606 | 4 port |
| 0.08785 | 24 port |
+-----+
9 rows in set (0.18 sec)

mysql>

```

This query analyzes the percentage of returns by product type (represented by product description) by joining the RMA and Orders tables. It groups the data by product description and calculates each product's share of total returns. The results show that the 48 port product type has the highest return percentage, followed by the 8 port product types, indicating that these products contribute the most to overall returns. This suggests certain product configurations may have higher defect rates or compatibility issues.



3. Write a report to clearly **summarize** your RMA **data analysis** for stakeholders. When you summarize the results, consider the following questions:
 - A. How does the data provide the product manager with usable information?
 - B. What are the potential flaws in the data that has been presented?
 - C. Are there any limitations on your conclusions or any other ways of looking at your findings that you haven't considered? Clearly communicate your findings to stakeholders.

The analysis of RMA data shows that return activity is concentrated in specific states and is heavily influenced by certain product types, particularly higher port configurations.

How does the data provide the product manager with usable information?

The data provides the product manager with usable information by identifying patterns in product returns across different states and product types. By analyzing the number of returns by state, the manager can see which regions have the highest return activity, such as Massachusetts, Arkansas, and Oregon. This helps highlight potential regional issues related to shipping, handling, or customer expectations. Additionally, analyzing the percentage of returns by product type shows which products contribute the most to overall returns. For example, products such as the 48 port and 8 port items have the highest return percentages. This information allows the product manager to focus on specific products and regions to improve quality, reduce return rates, and streamline operations.

What are the potential flaws in the data that has been presented?

There are several potential limitations in the data that may affect the accuracy of the analysis. One issue is that the dataset does not include clearly structured or standardized return reasons, making it difficult to determine why products are being returned. Another limitation is that product type is based on the description field, which may not be consistent or standardized across all records. This could lead to similar products being grouped separately or incorrectly. Additionally, the dataset may not capture all relevant factors, such as shipping conditions or customer satisfaction, which could influence return behavior.

Are there any limitations on your conclusions or any other ways of looking at your findings that you haven't considered? Clearly communicate your findings to stakeholders.

There are limitations to the conclusions that can be drawn from this analysis. The data does not include time-related information, so it is not possible to analyze trends over time or identify seasonal patterns in returns. It also does not provide insight into external factors such as shipping delays, product defects, or customer usage, which could impact return rates. Another limitation is that the analysis focuses only on returns and does not consider total sales volume, which would provide better context for understanding return rates. Other ways to expand this analysis could include examining return reasons in more detail, comparing return rates across different customer groups, or incorporating time-based data to identify trends. These additional approaches would provide a more complete understanding of return behavior and support better decision-making.